

Dropping things

Probing counterfactual thinking without counterfactual language

David Rose¹, Siying Zhang², Sophie Bridgers¹, Hyowon Gweon¹, and Tobias Gerstenberg¹

¹Stanford University, Department of Psychology

²University of Washington, Department of Psychology

Abstract

A hallmark of human intelligence is the ability to think about how things could have gone differently. Such counterfactual thinking underlies causal, social, and moral judgments. However, our understanding of how counterfactual thinking develops has been limited in at least two ways. First, many tasks rely on counterfactual language which may be difficult for young children to understand. Second, in most tasks, children can succeed without using genuine counterfactual thinking. We present a novel task that probes genuine counterfactual thinking without using counterfactual language. Data from both children ($N = 480$) and adults ($N = 90$) in the U.S. across three versions of the task suggest that genuine counterfactual thinking emerges around five years of age.

Keywords: counterfactual reasoning; hypothetical reasoning; development; social cognition.

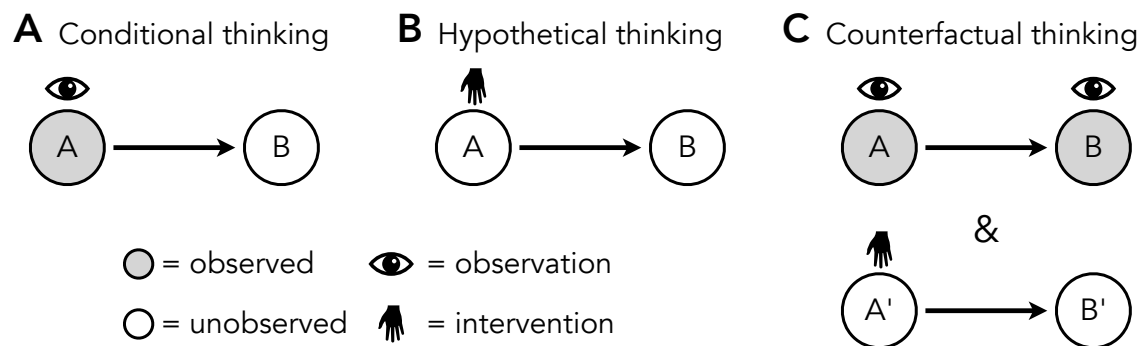
Introduction

David Lewis charged Hume with defining causation twice over: first, in terms of the regular succession of events, and second, in terms of *counterfactuals*—had the first event not occurred, then neither would the second (Lewis, 1973). What Hume identified, and Lewis clarified, is the central role that counterfactual thinking plays in determining what causes what. Yet, counterfactual thinking matters not just for reasoning about causation (German, 1999; Harris, German, & Mills, 1996; Koskuba, Gerstenberg, Gordon, Lagnado, & Schlottmann, 2018; Smallman & McCulloch, 2012; Summerville, 2011); it enables us to learn from past mistakes (Byrne, 2016; Roese, 1994), fosters creative problem solving (Kray, Galinsky, & Wong, 2006; Markman, Lindberg, Kray, & Galinsky, 2007), and even influences how we evaluate others’ actions and our own (Gerstenberg et al., 2018; Lagnado & Channon, 2008). For instance, when someone causes a bad outcome but could have avoided it, we hold them more responsible than when they couldn’t have done otherwise (Gautam & McAuliffe, 2024; Gautam, Owen Hall, Suddendorf, & Redshaw, 2023; Gerstenberg, 2024; Jaroslawska, McCormack, Burns, & Caruso, 2020; Koskuba et al., 2018; Kushnir, 2022; Lagnado, Gerstenberg, & Zultan, 2013; Payir & Guttentag, 2019; Pesowski, Denison, & Friedman, 2016; Stanley, Cabeza, Smallman, & De Brigard, 2021; Wong, Cordes, Harris, & Chernyak, 2023; Zhao, Zhao, Gweon, & Kushnir, 2021); when we ourselves cause a bad outcome, we experience regret when we could have avoided it (Byrne, 2016; Camille et al., 2004; Coricelli & Rustichini, 2010; Roese, 1994; Smallman & McCulloch, 2012; Summerville, 2011).

Despite the fundamental way in which counterfactual thinking shapes how we think, feel, and act, studying its development is challenging for at least two reasons. First, it is difficult to assess whether someone engaged in counterfactual thinking without having them respond to verbal prompts about counterfactuals. For example, after having seen a scenario where the wind causes leaves to fall from a tree, children may be asked: “Would the leaves have fallen from the tree if there had been no wind?” (Harris, 1997, see also Nyhout & Ganea, 2019, 2020; Nyhout, Henke, & Ganea, 2017; Rafetseder, Cristi-Vargas, & Perner, 2010). Correctly parsing such complex sentences is challenging, sometimes even for adults (see e.g., Tulling, Bacon, & Cournane, 2025). Thus, the linguistic demands may mask existing counterfactual thinking abilities especially in young children.

Second, it is difficult to distinguish counterfactual thinking from related forms of thinking, such as conditional or hypothetical thinking (see Figure 1). For instance, to correctly answer that the leaves would not have fallen from the tree if there had been no wind, one can rely on conditional thinking from cause to effect without invoking counterfactuals (Beck & Guthrie, 2011; Rafetseder et al., 2010). To probe genuine counterfactual thinking, it is critical to use tasks that require taking into account what actually happened, mentally traveling back in time to imagine a change to an event, and then simulating how this alternative scenario would have played out (Gerstenberg, 2022).

To address these challenges, we need a task that probes genuine counterfactual thinking without relying on counterfactual language. While asking whether children experience regret (e.g., Amsel & Smalley, 2014; Beck & Crilly, 2009; Guttentag & Ferrell, 2004; Jones, Nelson, Gautam, & Redshaw, 2025; McCormack, O’Connor, Beck, & Feeney, 2016) provides one way to probe genuine counterfactual thinking without counterfactual

**Figure 1**

Conditional, Hypothetical, and Counterfactual Thinking: **A** Conditional thinking involves drawing inferences about unobserved events from observed events. Note that one can reason from (observed) cause to (unobserved) effect (e.g., predicting where a basketball will land on the floor), or from effect to cause (e.g., inferring where a basketball on the floor must have come from). **B** Hypothetical thinking involves considering the consequences of actions (e.g., predicting what would happen if a basketball were dropped). **C** Counterfactual thinking involves considering how things could have played out differently (e.g., considering what would have happened if the basketball had been dropped but not caught, when in fact it was caught). While both hypothetical thinking and counterfactual thinking involve considering the consequences of interventions, the key difference is whether what actually happened was observed. Hypothetical thinking considers interventions in the future, whereas counterfactual thinking considers interventions in the past.

language, this approach requires children to identify, label, and report their internal affective experiences, which may also develop throughout early childhood (e.g., Jones, Gautam, & Redshaw, 2024; McCormack, Feeney, & Beck, 2020; Rafetseder & Perner, 2012; Weisberg & Beck, 2012). Here, we take a novel approach to probing genuine counterfactual thinking without counterfactual language: Leveraging children’s early-developing understanding of intuitive physics (e.g., Hespos & vanMarle, 2011; Piloto, Weinstein, Battaglia, & Botvinick, 2022; Spelke, Breinlinger, Macomber, & Jacobson, 1992) and prosocial actions (e.g., Brownell, 2012; Hepach & Warneken, 2018; Noles & McDermott, 2023; Vaish & Savell, 2022).

In this task, participants watch a character (Granny) drop two items—each of which is caught by two different characters—and answer a question about who should receive a thank-you sticker. This task does not require counterfactual language, in either comprehension or production; participants simply need to understand that only one character can receive a sticker and choose who should get it. To distinguish counterfactual thinking from hypothetical and conditional thinking (see Figure 1), we tested children and adults in three different versions of the task (see Figure 2). All experiments were pre-registered. The materials, data, pre-registrations, and analyses are available at https://anonymous.4open.science/r/counterfactual_development_anon-3111.

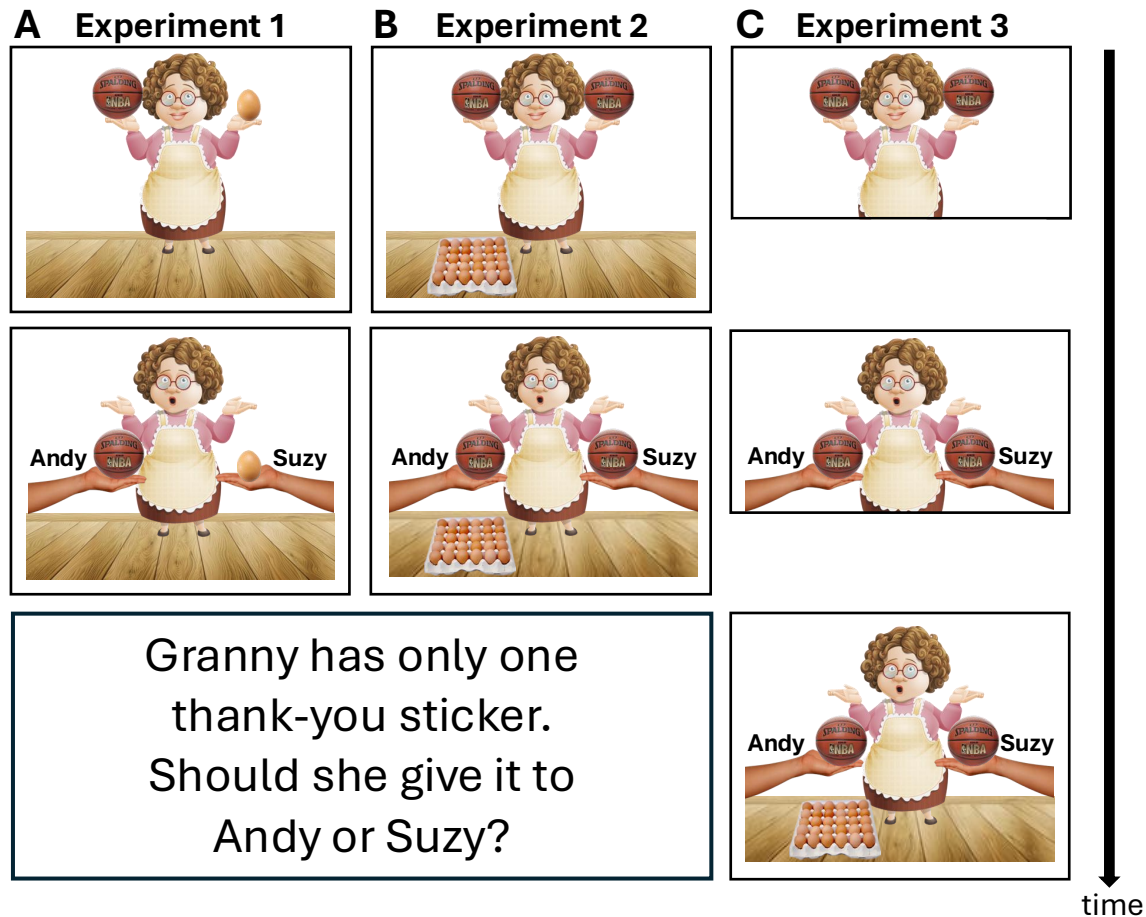


Figure 2

Experiment Overview: In all experiments, Granny drops two objects, and each of them is caught. She only has one thank-you sticker. Who should she give it to? **A** In Experiment 1, Granny drops two different objects. It would have been worse if the egg hadn't been caught than if the basketball hadn't been caught. **B** In Experiment 2, Granny drops two objects of the same kind (e.g., two basketballs). Here, one of the objects would have landed on a carton of eggs and the other on the floor. **C** In Experiment 3, the full scene is revealed only after two objects of the same kind were caught. This means that participants cannot anticipate early on what would happen if Granny dropped the objects.

Experiment 1: Counterfactual thinking without counterfactual language

Methods and results

Participants

Our final sample included 120 children, who were between the ages of 3 and 6, recruited through Lookit who met our pre-registered inclusion criteria (*gender*: 60 female, 59 male, 1 no response/other; *language*: 114 English, 6 no response/other). 82 children

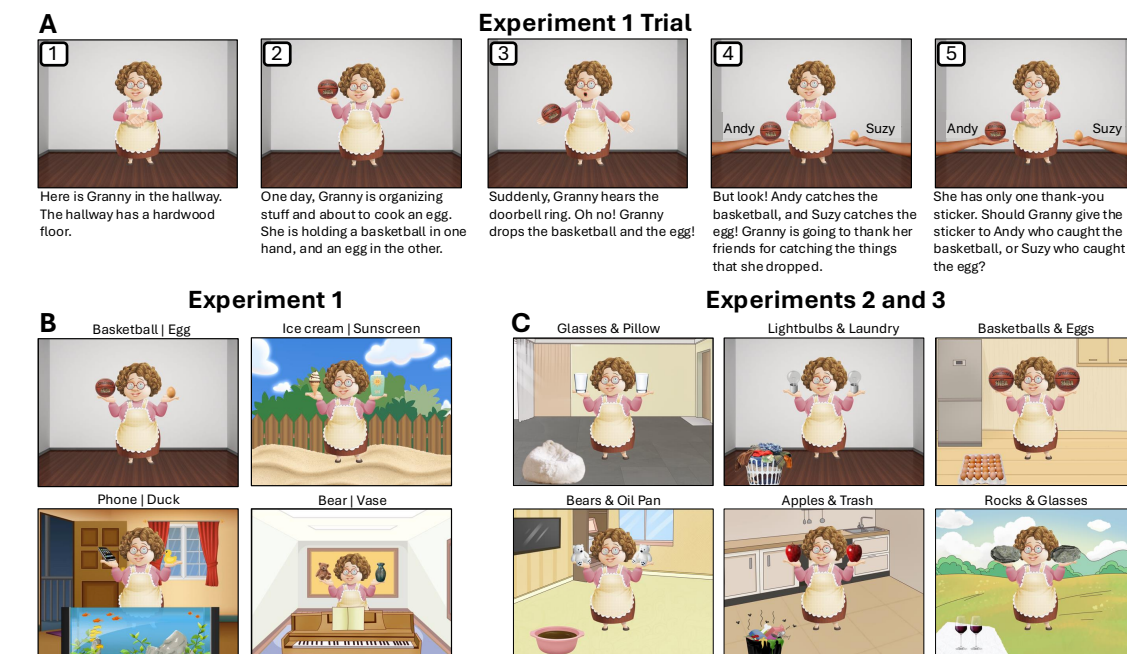


Figure 3

A *Experiment 1 example trial:* 1 Participants first see that Granny is in a hallway. And they are told that the hallway has hardwood floors. 2 Granny is holding a basketball and egg. 3 Granny drops the objects after being startled by the doorbell. 4 The two objects are caught by Andy and Suzy. 5 Participants are asked who Granny should give the thank-you sticker to. **Experiment scenarios:** Each panel in B and C shows a scene (#2) from scenarios that participants saw in each experiment. **B** In Experiment 1, participants saw four scenarios where Granny drops two different objects. **C** In Experiments 2 and 3, participants saw six scenarios where Granny drops two of the same objects.

were excluded for failing the warmup trial. To be included, participants needed to pass the warmup trial. Since our task involved a number of friends' names and objects across trials, we wanted to ensure that participants could track this. To accommodate younger participants, we allowed children to respond by pointing, with parents verbally reporting their child's answer. In these cases, we ensured parents remained neutral and did not influence their child's responses. Despite these accommodations, many 3-year-olds still could not meet this basic comprehension criterion and were therefore excluded from the main analysis. Each age group included 30 participants. Families received \$5. Additionally, we recruited 30 adult participants through Prolific who met our pre-registered inclusion criteria (*age*: $M = 35$, $SD = 13$; *gender*: 21 female, 9 male; *race*: 21 White, 5 no response/other, 3 Black, 1 Asian). Participants were compensated at a rate of \$12 per hour.

Materials

We created six trials, where four of these were test trials, one was a warm-up and one was a post-test trial. The warm-up trial showed Granny, a plant and a lamp on opposite

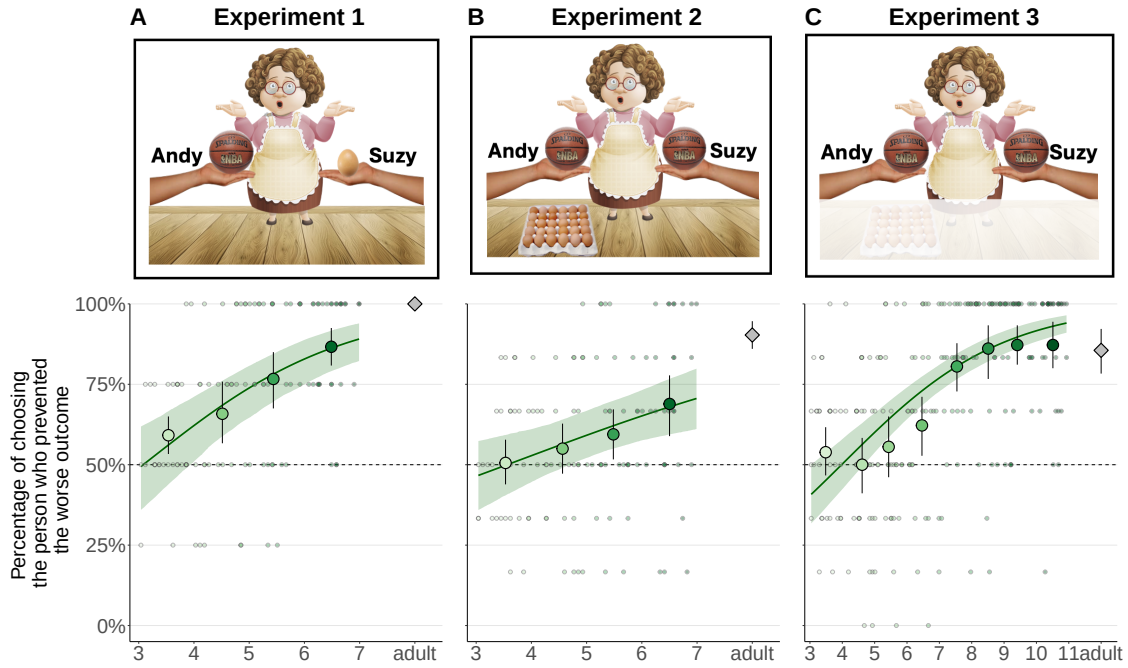


Figure 4

Overall Experimental Results: Percentage of participants who chose the person who prevented the worse outcome from happening across the three experiments. Lines show the best fit from a Bayesian logistic regression model. The ribbon shows the 95% credible interval for the regression line. Large points show means for each age group. Error bars show 95% bootstrapped confidence intervals. Small points show individual responses averaged across trials. Dashed lines indicate chance performance. Each experiment had 30 participants in each age group, except for Experiment 2 which had 31 adults. **A** In Experiment 1, two different objects are dropped and prevented from falling onto the ground. **B** In Experiment 2, two of the same kinds of objects are dropped and prevented from falling. **C** In Experiment 3, participants saw where the objects would have landed only after they were caught. Initially, only Granny and the two identical objects were visible. After the objects were caught, the screen zoomed out so that participants could see the entire scene.

sides of her, and two hands, with one by the plant and one by the lamp. The materials for the test trials depicted Granny holding two different objects in different situations. These are shown in Figure 3b. The post-test trial showed Granny outside by two piles of leaves, where one pile was smaller than the other.

Procedure

The experiment for children was programmed using Lookit (Scott & Schulz, 2017) and for adults it was programmed using jsPsych (De Leeuw, 2015). Both children and adults were tested asynchronously.

Participants began with the warm-up trial. They were first introduced to Granny.

They were then told that Granny had a plant and a lamp. Each of these was on opposite sides of Granny and both wiggled when they were introduced. Then participants were told that they would meet two of Granny’s friends. They were shown an arm on the plant side of the screen, told that this is Benji and that Benji is by the plant. And they were shown an arm on the lamp side of the screen, told that this is Harry and that Harry is by the lamp. Participants were then asked who is by the plant and who is by the lamp. Children responded out loud. Adults clicked one of two buttons that were labeled “Benji” and “Harry”.

Participants then proceeded to the four test trials, where each involved Granny in different situations, holding different objects and dropping them. Both objects are then caught. [Figure 3a](#) shows an example trial and [Figure 3b](#) shows the different scenarios.

In all test trials, after the two objects were dropped and caught, participants were told that Granny only has one thank you sticker. Participants were asked who Granny should give it to. Children responded out loud and adults selected one of two buttons with the name of the individuals who caught the objects.

Lastly, participants completed a post-test trial. They were told that Granny has two of her friends help her to rake leaves. They are then shown two piles of leaves, one of which was much smaller than the other, and asked who Granny should thank. The order of the test trials was randomized within participants. In addition, for all test trials, the side of the screen that the dropped objects were on and names of the individuals who caught the objects was randomized across participants. The position of the object that would result in a worse outcome was counterbalanced across the four test trials—with two being on the left side and two being on the right side.

Results

We pre-registered two hypotheses for children. First, that they would be more likely to say that the individual who prevented the worse potential outcome from happening should get the thank-you sticker. Second, that the likelihood of choosing the person who prevented the worse potential outcome would increase with age. For adults, we pre-registered the hypothesis that they would be more likely to choose the individual who prevented the worse potential outcome.

The results are shown in [Figure 4a](#). For all experiments, we coded responses so that 1 = selecting the person who prevented the worse outcome and 0 = selecting the other person. As predicted, we found that children were more likely to select the individual who prevented the worse outcome from happening as the person who should receive the thank-you sticker, $\beta = 0.73$, 95% credible interval (CrI) = [0.65, 0.80]. Older children were more likely to select the individual who prevented the worse outcome from happening than younger children, $\beta = 0.55$, 95% CrI = [0.34, 0.75]. The estimated age at which children exceeded chance performance was 3.95 years. Adults chose the individual who prevented the worse outcome 100% of the time.

Discussion

When two different objects are dropped and caught, children showed above-chance performance by around 4 years of age, correctly choosing the person who prevented the

worse outcome as the one who should receive the thank-you sticker. This result raises the possibility that in our study, by around 4 years of age, children engaged in counterfactual thinking to decide whom Granny should thank. That is, they may have inferred what would have happened if the characters had not caught the objects, and chosen the one who prevented the worse outcome from happening. However, given that each scenario used in Experiment 1 involved dropping two different objects, we cannot rule out the possibility that children merely considered which object Granny (or they themselves) might like more and chose the person who caught the preferred object (see [Figure B1](#) for relevant empirical data).

Experiment 2 addresses this concern. In this experiment, Granny drops two objects of the same kind (see [Figure 3c](#)). This means that children cannot give the correct answer by considering which object Granny or they themselves might like better.

Experiment 2: Addressing the preference-based account

Methods and results

Participants

Our final sample included 120 children, who were between the ages of 3 and 6, recruited through Lookit who met our pre-registered inclusion criteria (*gender*: 65 female, 55 male; *language*: 118 English, 2 no response/other). 74 participants were excluded for failing to pass the warmup trial. Each age group included 30 participants. Families received \$5. Additionally, we recruited 31 adult participants through Prolific who met our pre-registered inclusion criteria (*age*: $M = 39$, $SD = 14$; *gender*: 19 female, 11 male, 1 no response/other; *race*: 20 White, 7 Black, 2 no response/other, 1 Asian). Participants were compensated at a rate of \$12 per hour.

Materials

The warm-up trial was the same as in Experiment 1. The materials for the test trials, however, differed from Experiment 1 in that they depicted Granny holding two of the same objects in different situations. All six test trials are shown in [Figure 3c](#).

Procedure

The procedure and design were the same as in Experiment 1. An example trial is shown in [Figure C1](#).

Results

The pre-registered hypotheses were the same as in Experiment 1. The results are shown in [Figure 4b](#). As predicted, we found that both children and adults were more likely to select the individual who prevented the worse outcome from happening as the person who should receive the thank-you sticker (children: $\beta = 0.37$, 95% CrI = [0.07, 0.72], adults: $\beta = 2.44$, 95% CrI = [1.49, 3.44]). Older children were more likely to select the individual who prevented the worse outcome from happening than younger children, $\beta = 0.26$, 95% CrI = [0.09, 0.44]. The estimated age at which children exceeded chance performance was 4.71 years.

Discussion

When two of the same objects are dropped and caught, children showed above-chance performance by around 5 years of age, correctly choosing the person who prevented the worse outcome as the one who should receive the thank-you sticker. Compared to Experiment 1 where two different objects were dropped onto the same surface, children’s performance in Experiment 2 was lower. Along with data in [Figure B1](#), this suggests that the strong performance in Experiment 1 can at least be partially explained by children’s tendency to choose the person who caught the object that they themselves preferred (or they thought Granny might prefer). This preference-based account, however, does not apply to Experiment 2.¹ Here, children need to simulate what would happen if the objects were not caught, determine which outcome would be worse, and then select the person who prevented the worse outcome as the one who should receive the thank-you sticker.

There is, however, a remaining concern. In Experiment 2, participants saw the whole scene from the beginning. This raises the possibility that they succeeded by engaging in hypothetical thinking, merely simulating “what would happen” if Granny dropped these objects prior to seeing the actual events unfold in the scenario. For example, when Granny holds a basketball over a carton of eggs and another basketball over the floor, they can simulate that dropping the basketball over the egg carton would be worse than dropping the basketball over the floor. They can then compare the outcome of these hypothetical simulations to what actually happened, and make the judgment that catching the basketball over the egg carton is more deserving of a reward.

Experiment 3 addresses this concern. It is similar to Experiment 2, except that the full scene is revealed only *after* the objects were caught (see [Figure 4c](#)). This subtle difference has a powerful effect: it renders hypothetical thinking ineffective because it’s impossible to anticipate which of the two objects is more important to catch. As such, Experiment 3 provides a stronger test for genuine counterfactual thinking.

Experiment 3: Genuine counterfactual thinking

Methods and results

Participants

We expanded the age range from previous experiments since we expected the task to be more challenging. Our final sample included 240 children, aged 3 to 10, recruited through Lookit who met our pre-registered inclusion criteria (*gender*: 137 female, 102 male, 1 no response/other; *language*: 235 English, 5 no response/other). 101 participants were excluded for failing to pass the warmup trial. Each age group included 30 participants. Families received \$5. Additionally, we recruited 30 adult participants through Prolific who met our pre-registered inclusion criteria (*age*: $M = 39$, $SD = 12$; *gender*: 20 female, 9 male, 1 no response/other; *race*: 23 White, 5 Black, 1 no response/other, 1 Asian).

¹While it is in principle possible that some children preferred one type of surface over another, it is unlikely that these preferences would be systematic.

Materials

The warm-up trial was the same as in our previous experiments. The materials were the same as in Experiment 2.

Procedure

The procedure and design were the same as in Experiment 2 except that it was revealed only after the objects were caught, where they would have fallen. An example trial is shown in [Figure D1](#).

Results

Our pre-registered hypotheses were the same as in Experiments 1 and 2. The results are shown in [Figure 4c](#). As predicted, we found that both children and adults were more likely to select the individual who prevented the worse outcome from happening as the person who should receive the thank-you sticker (children: $\beta = 1.19$, 95% CrI = [0.95, 1.45], adults: $\beta = 2.78$, 95% CrI = [0.74, 4.99]). Older children were more likely to select the individual who prevented the worse outcome from happening than younger children, $\beta = 0.40$, 95% CrI = [0.31, 0.50]. The estimated age at which children exceeded chance performance was 5.02 years.²

Discussion

When two of the same objects were dropped and caught, and the potential landing spots were revealed only later, children showed above-chance performance by around 5 years of age. Moreover, accuracy increased substantially by around age 7, nearing ceiling and adult performance. Unlike in Experiment 2, where participants could run hypothetical simulations before the objects were dropped, this strategy was unavailable here. Instead, participants needed to engage in genuine counterfactual thinking to decide who should receive the thank-you sticker.

General Discussion

Counterfactual thinking is a fundamental cognitive capacity. Yet, probing the development of genuine counterfactual thinking is challenging. Some tasks may have masked counterfactual thinking abilities by using counterfactual language which is difficult to process. Other tasks may have failed to assess genuine counterfactual thinking which requires going back in time, imagining a change to what happened, and simulating how things would have turned out differently (see [Figure 1](#)). Using a novel task that addresses both challenges we explored the development of counterfactual reasoning.

In the ‘dropping things’ task, two objects are dropped and caught; however, the consequences of one object completing its fall would have been worse than the other. For example, if two basketballs are dropped and caught, but one would have fallen on a carton

²For this analysis, we restricted the age range from 3- to 6-year-old children, as in Experiments 1 and 2. Using the full age range, we find that children perform above chance at 4.67 years old. This shift is due to the high performance of older children. We would expect a similar shift for Experiments 1 and 2, if the same age range was tested as in Experiment 3.

of eggs and the other on the floor, catching the basketball above the eggs prevented a worse outcome. To appreciate this, one needs to take into account what actually happened (the balls were caught), mentally travel back in time to imagine a change to an event (the balls were not caught), and then simulate how this alternative scenario would have played out (one ball would break a carton of eggs while the other would bounce off the floor). To probe whether children appreciate this, we asked them to whom Granny (who dropped the objects) should give her only thank-you sticker. In this way, the ‘dropping things’ task probes genuine counterfactual thinking without using counterfactual language.

Tasks that involve regret also probe counterfactual thinking without using counterfactual language (e.g., [Amsel & Smalley, 2014](#); [Beck & Crilly, 2009](#); [Guttentag & Ferrell, 2004](#); [Jones et al., 2025](#); [McCormack et al., 2020, 2016](#); [O’Connor, McCormack, & Feeney, 2012](#); [Rafetseder & Perner, 2012](#); [Weisberg & Beck, 2012](#)). For example, when a child chooses one of two boxes, rates their happiness with the outcome, and later finds out that the other box contained a better prize, they often rate themselves as less happy ([Amsel & Smalley, 2014](#)). This change in happiness is thought to reflect counterfactual thinking—realizing that an alternative decision would have yielded a better outcome (e.g., [Byrne, 2016](#); [Coricelli & Rustichini, 2010](#)).

Yet, there has been substantial variability in when children begin to succeed in tasks that rely on regret ([Jones et al., 2024](#); [McCormack et al., 2020](#); [Rafetseder & Perner, 2012](#); [Weisberg & Beck, 2012](#)); while some research suggests that they do so reliably by age 6 (see e.g., [McCormack et al., 2020](#)), others have found much later success (see e.g., [Rafetseder & Perner, 2012](#)). Note that experiencing regret requires an affective appraisal on top of counterfactual thinking ([Beck & Crilly, 2009](#); [McCormack et al., 2020](#); see [Jones et al., 2024](#) for a discussion of additional limitations). It is thus possible that regret tasks underestimate the age at which counterfactual thinking develops. The ‘dropping things’ task doesn’t require a first-person affective appraisal; instead, it leverages children’s intuitive, early-emerging understanding of the physical and social world (see e.g., [Brownell, 2012](#); [Hepach & Warneken, 2018](#); [Hespos & vanMarle, 2011](#); [Spelke et al., 1992](#)). Even young children have plenty of experience with objects being dropped, either by themselves or by adults—often followed by frustration. And saying “thank you” or receiving a sticker for doing something good is a social act that children learn early on (e.g., [Noles & McDermott, 2023](#); [Vaish & Savell, 2022](#)).

Still, one might wonder whether the ‘dropping things’ task has fully ruled out the possibility that participants succeed via hypothetical, rather than counterfactual thinking. In Experiment 3, participants cannot run hypothetical simulations before the objects are dropped. Nonetheless, they could in principle run such simulations after knowing what happened. At that point, however, the distinction between hypothetical and counterfactual simulation becomes blurry. If we knew everything that happened in advance, then a hypothetical simulation would yield the same result as a counterfactual simulation (because there is no more benefit of hindsight). These two forms of thinking only come apart when there is some uncertainty about the future that gets resolved based on what happened ([Gerstenberg, 2022](#)). Our experiment was designed precisely to create this resolution point. In doing so, it ensures that participants were not just imagining possible futures, but were actively contrasting what in fact occurred with what could have happened otherwise—the hallmark of counterfactual thought.

Conclusion

Counterfactual thinking is a critical human capacity. We present the ‘dropping things’ task—a new task for studying counterfactual thinking. This task doesn’t require counterfactual language, and tests genuine counterfactual thinking. The task can easily be adapted and administered in other languages across different platforms and cultural contexts. Much like how the Sally-Anne task provided a novel way to study the development of children’s understanding of mental states, in a way that that revolutionized research on theory of mind (Baron-Cohen, Leslie, & Frith, 1985; Dennett, 1978; Wimmer & Perner, 1983), we believe that the ‘dropping things’ task has a similar potential to dramatically improve our understanding of how counterfactual thinking develops. Better understanding the developmental onset and variability of counterfactual thinking will help us better characterize how early cognition can function without it and how, once acquired, this capacity transforms children’s reasoning about the world around them.

References

- Amsel, E., & Smalley, J. D. (2014). Beyond really and truly: Children's counterfactual thinking about pretend and possible worlds. In *Children's reasoning and the mind* (pp. 121–147). Psychology Press.
- Baron-Cohen, S., Leslie, A. M., & Frith, U. (1985). Does the autistic child have a “theory of mind”? *Cognition*, 21(1), 37–46.
- Beck, S. R., & Crilly, M. (2009). Is understanding regret dependent on developments in counterfactual thinking? *British Journal of Developmental Psychology*, 27(2), 505–510.
- Beck, S. R., & Guthrie, C. (2011). Almost thinking counterfactually: Children's understanding of close counterfactuals. *Child development*, 82(4), 1189–1198.
- Brownell, C. A. (2012). Early development of prosocial behavior: Current perspectives. *Infancy*, 18(1), 1–9. doi: 10.1111/inf.12004
- Byrne, R. M. (2016). Counterfactual thought. *Annual Review of Psychology*, 67, 135–157.
- Camille, N., Coricelli, G., Sallet, J., Pradat-Diehl, P., Duhamel, J.-R., & Sirigu, A. (2004). The involvement of the orbitofrontal cortex in the experience of regret. *Science*, 304(5674), 1167–1170.
- Coricelli, G., & Rustichini, A. (2010). Counterfactual thinking and emotions: regret and envy learning. *Philosophical Transactions of the Royal society B: Biological sciences*, 365(1538), 241–247.
- De Leeuw, J. R. (2015). jspsych: A javascript library for creating behavioral experiments in a web browser. *Behavior research methods*, 47, 1–12.
- Dennett, D. C. (1978). Beliefs about beliefs. *Behavioral and Brain sciences*, 1(4), 568–570.
- Gautam, S., & McAuliffe, K. (2024). Why imagining what could have happened matters for children's social cognition. *WIREs: Cognitive Science*, 15(1).
- Gautam, S., Owen Hall, R., Suddendorf, T., & Redshaw, J. (2023). Counterfactual choices and moral judgments in children. *Child Development*, 94(5), e296–e307.
- German, T. P. (1999). Children's causal reasoning: Counterfactual thinking occurs for ‘negative’ outcomes only. *Developmental Science*, 2(4), 442–457.
- Gerstenberg, T. (2022). What would have happened? counterfactuals, hypotheticals and causal judgements. *Philosophical Transactions of the Royal Society B*, 377(1866), 20210339.
- Gerstenberg, T. (2024). Counterfactual simulation in causal cognition. *Trends in Cognitive Sciences*, 28(10), 924–936.
- Gerstenberg, T., Ullman, T. D., Nagel, J., Kleiman-Weiner, M., Lagnado, D. A., & Tenenbaum, J. B. (2018). Lucky or clever? From expectations to responsibility judgments. *Cognition*, 177, 122–141. doi: 10.1016/j.cognition.2018.03.019
- Guttentag, R., & Ferrell, J. (2004). Reality compared with its alternatives: age differences in judgments of regret and relief. *Developmental Psychology*, 40(5), 764.
- Harris, P. (1997). On realizing what might have happened instead. *Polish Quarterly of Developmental Psychology*, 3, 161–176.
- Harris, P., German, T., & Mills, P. (1996). Children's use of counterfactual thinking in causal reasoning. *Cognition*, 61(3), 233–259.
- Hepach, R., & Warneken, F. (2018). Editorial overview: Early development of proso-

- cial behavior: Revealing the foundation of human prosociality. *Current Opinion in Psychology*, 20, iv–viii. doi: 10.1016/j.copsyc.2018.02.001
- Hespos, S. J., & vanMarle, K. (2011). Physics for infants: characterizing the origins of knowledge about objects, substances, and number. *Wiley Interdisciplinary Reviews: Cognitive Science*, 3(1), 19–27. doi: 10.1002/wcs.157
- Jaroslawska, A. J., McCormack, T., Burns, P., & Caruso, E. M. (2020). Outcomes versus intentions in fairness-related decision making: School-aged children's decisions are just like those of adults. *Journal of experimental child psychology*, 189, 104704.
- Jones, A. K., Gautam, S., & Redshaw, J. (2024). Young children experience both regret and relief in a gain-or-loss context. *Cognition and Emotion*, 38(1), 163–170.
- Jones, A. K., Nelson, N. L., Gautam, S., & Redshaw, J. (2025). Children infer counterfactual information from others' facial expressions. *Journal of Experimental Child Psychology*, 259, 106306.
- Koskuba, K., Gerstenberg, T., Gordon, H., Lagnado, D. A., & Schlottmann, A. (2018). What's fair? how children assign reward to members of teams with differing causal structures. *Cognition*, 177, 234–248.
- Kray, L. J., Galinsky, A. D., & Wong, E. M. (2006). Thinking within the box: The relational processing style elicited by counterfactual mind-sets. *Journal of personality and social psychology*, 91(1), 33.
- Kushnir, T. (2022). Imagination and social cognition in childhood. *Wiley Interdisciplinary Reviews: Cognitive Science*, 13(4), e1603.
- Lagnado, D. A., & Channon, S. (2008). Judgments of cause and blame: The effects of intentionality and foreseeability. *Cognition*, 108(3), 754–770.
- Lagnado, D. A., Gerstenberg, T., & Zultan, R. (2013). Causal responsibility and counterfactuals. *Cognitive Science*, 47, 1036–1073.
- Lewis, D. (1973). Causation. *The Journal of Philosophy*, 70(17), 556–567.
- Markman, K. D., Lindberg, M. J., Kray, L. J., & Galinsky, A. D. (2007). Implications of counterfactual structure for creative generation and analytical problem solving. *Personality and Social Psychology Bulletin*, 33(3), 312–324.
- McCormack, T., Feeney, A., & Beck, S. R. (2020). Regret and decision-making: A developmental perspective. *Current Directions in Psychological Science*, 29(4), 346–350.
- McCormack, T., O'Connor, E., Beck, S., & Feeney, A. (2016). The development of regret and relief about the outcomes of risky decisions. *Journal of Experimental Child Psychology*, 148, 1–19. doi: 10.1016/j.jecp.2016.02.008
- Noles, N. S., & McDermott, C. H. (2023). Children's understanding of gratitude, generosity, and reciprocity. *Cognitive Development*, 66, 101322.
- Nyhout, A., & Ganea, P. A. (2019). Mature counterfactual reasoning in 4- and 5-year-olds. *Cognition*, 183, 57–66. doi: 10.1016/j.cognition.2018.10.027
- Nyhout, A., & Ganea, P. A. (2020). What is and what never should have been: Children's causal and counterfactual judgments about the same events. *Journal of Experimental Child Psychology*, 104773. Retrieved from <http://dx.doi.org/10.1016/j.jecp.2019.104773> doi: 10.1016/j.jecp.2019.104773
- Nyhout, A., Henke, L., & Ganea, P. A. (2017). Children's counterfactual reasoning about causally overdetermined events. *Child Development*. doi: 10.1111/cdev.12913
- O'Connor, E., McCormack, T., & Feeney, A. (2012). The development of regret. *Journal*

- of *experimental child psychology*, 111(1), 120–127.
- Payir, A., & Guttentag, R. (2019). Counterfactual thinking and age differences in judgments of regret and blame. *Journal of Experimental Child Psychology*, 183, 261–275.
- Pesowski, M. L., Denison, S., & Friedman, O. (2016). Young children infer preferences from a single action, but not if it is constrained. *Cognition*, 155, 168–175.
- Piloto, L. S., Weinstein, A., Battaglia, P., & Botvinick, M. (2022). Intuitive physics learning in a deep-learning model inspired by developmental psychology. *Nature Human Behaviour*, 6(9), 1257–1267. doi: 10.1038/s41562-022-01394-8
- Rafetseder, E., Cristi-Vargas, R., & Perner, J. (2010). Counterfactual reasoning: Developing a sense of “nearest possible world”. *Child development*, 81(1), 376–389.
- Rafetseder, E., & Perner, J. (2012). When the alternative would have been better: Counterfactual reasoning and the emergence of regret. *Cognition & emotion*, 26(5), 800–819.
- Roe, N. J. (1994). The functional basis of counterfactual thinking. *Journal of personality and Social Psychology*, 66(5), 805.
- Scott, K., & Schulz, L. (2017). Lookit (part 1): A new online platform for developmental research. *Open Mind*, 1(1), 4–14.
- Smallman, R., & McCulloch, K. C. (2012). Learning from yesterday’s mistakes to fix tomorrow’s problems: When functional counterfactual thinking and psychological distance collide. *European journal of social psychology*, 42(3), 383–390.
- Spelke, E. S., Breinlinger, K., Macomber, J., & Jacobson, K. (1992). Origins of knowledge. *Psychological review*, 99(4), 605.
- Stanley, M. L., Cabeza, R., Smallman, R., & De Brigard, F. (2021). Memory and counterfactual simulations for past wrongdoings foster moral learning and improvement. *Cognitive Science*, 45(6), e13007.
- Summerville, A. (2011). Counterfactual seeking: The scenic overlook of the road not taken. *Personality and Social Psychology Bulletin*, 37(11), 1522–1533.
- Tulling, M. A., Bacon, M., & Cournane, A. (2025). The past is “fake”: Facilitated processing of wishes compared with counterfactual conditionals in 4- and 5-year-olds. *Journal of experimental child psychology*, 255, 106220.
- Vaish, A., & Savell, S. (2022). Young children value recipients who display gratitude. *Developmental Psychology*, 58(4), 680.
- Weisberg, D. P., & Beck, S. R. (2012). The development of children’s regret and relief. *Cognition & emotion*, 26(5), 820–835.
- Wimmer, H., & Perner, J. (1983). Beliefs about beliefs: Representation and constraining function of wrong beliefs in young children’s understanding of deception. *Cognition*, 13(1), 103–128.
- Wong, A., Cordes, S., Harris, P. L., & Chernyak, N. (2023). Being nice by choice: The effect of counterfactual reasoning on children’s social evaluations. *Developmental Science*, 26(6), e13394.
- Zhao, X., Zhao, X., Gweon, H., & Kushnir, T. (2021). Leaving a choice for others: Children’s evaluations of considerate, socially-mindful actions. *Child development*, 92(4), 1238–1253.

Supplemental Materials A

Experiment 1

Trial results

The results for each of the four trials are shown in Figure A1. As can be seen, they are largely consistent across trials.

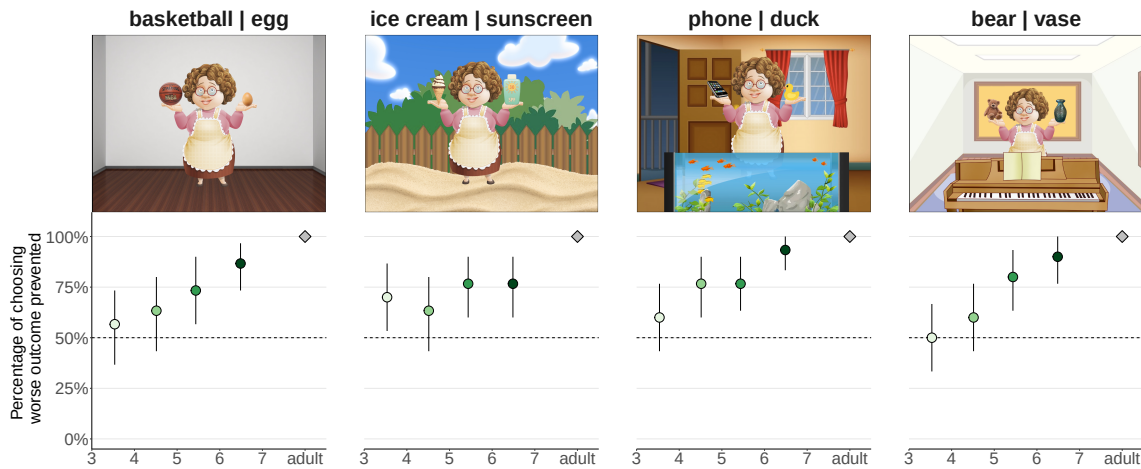


Figure A1

Experiment 1: Percentage of participants who chose the person who prevented the worse outcome from happening in each of the four test trials. Points show means for each age group with 95% bootstrapped confidence intervals. Dashed lines indicate chance performance. The pattern of performance was similar across the four different scenarios.

Supplemental Materials B

Experiment 1: Which object do you like? Which object does Granny like?

In this experiment, we asked whether children might select who to thank by thinking about what object they like or what object Granny likes.

Participants

Our final sample included 120 children, who were between the ages of 3 and 6, through Lookit who met our pre-registered inclusion criteria (*gender*: 57 female, 63 male; *language*: 117 English, 3 no response/other). Each age group included 30 participants. Families received \$5.

Materials

The materials were the same as in Experiment 1.

Procedure

Children completed two blocks that included four trials each. In each block, Granny is holding the same pairs of objects as in Experiment 1. The setup within each block is

identical, featuring the same content but what question children were asked to answer differed between each block. In one block, children were asked “Which one do you like more?”. In the other, they were asked “Which one do you think Granny likes more?”. In both blocks, we randomized on what side each object was shown. The order of test trials within each block was also randomized. Lastly, the presentation order of the two blocks was counterbalanced.

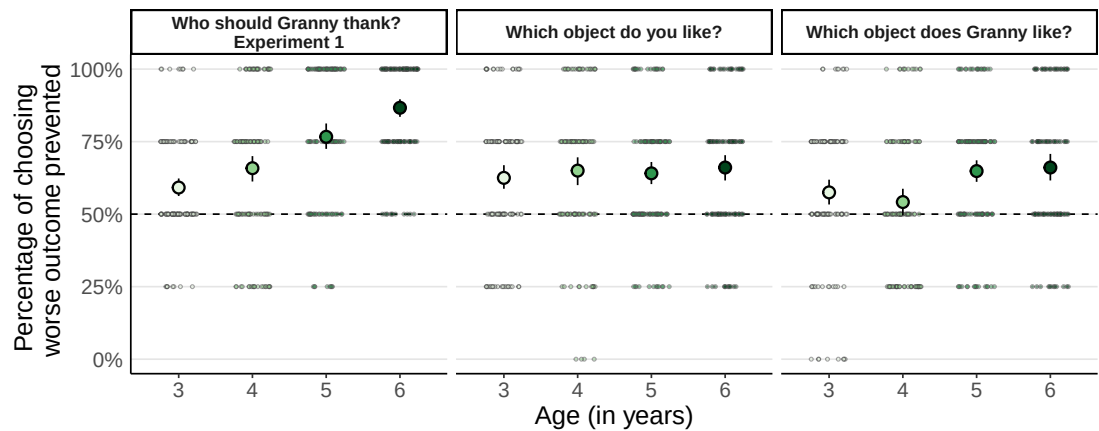


Figure B1

Experiment 1: Which object do you like? Which object does Granny like?: Percentage of participants who chose the person who prevented the worse outcome from happening. Experiment 1 results are shown on the left for comparison. Large points show means for each age group with 95% bootstrapped confidence intervals. Small points show individual responses averaged across trials. Dashed lines indicate chance performance.

Results

The results are shown in [Figure B1](#). As can be seen, children’s performance is similar in the preference tasks to the thank-you sticker task, but the age trend is stronger for the thank-you sticker task.

Supplemental Materials C
Experiment 2

Example trial

[Figure C1](#) shows an example trial from Experiment 2.

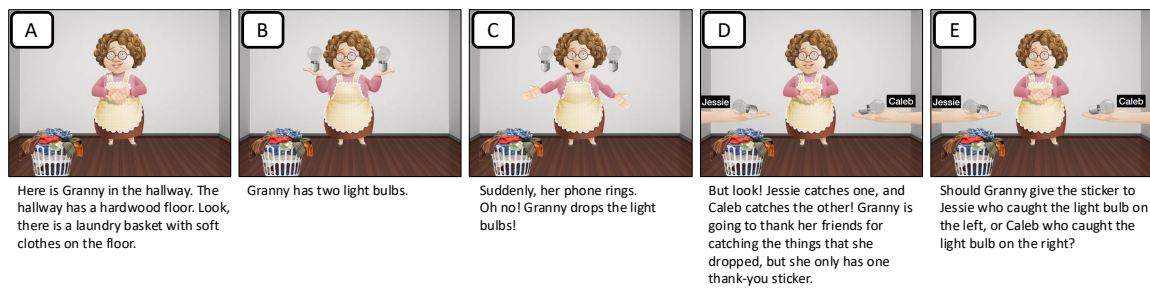


Figure C1

Experiment 2 example trial: Each panel shows what participants saw and what they were told. **A** Participants first see that Granny is in a hallway. And they are told that the hallway has hardwood floors, and that a laundry basket is on the floor. **B** Granny is holding two lightbulbs. **C** Granny drops the objects after being startled by the phone ringing. **D** The two objects are caught by Jessie and Caleb. **E** Participants are asked who Granny should thank.

Trial results

The results for each of the six trials are shown in Figure C2. As can be seen, they are largely consistent across trials.

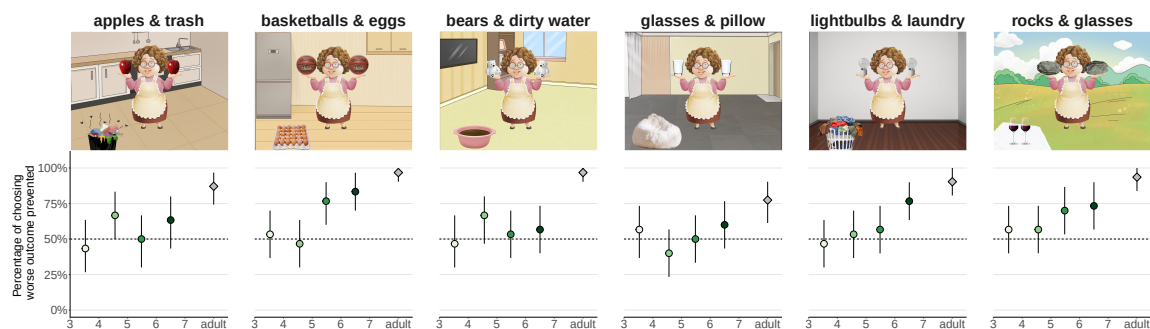


Figure C2

Experiment 2: Percentage of participants who chose the person who prevented the worse outcome from happening in each of the six test trials. Points show means for each age group with 95% bootstrapped confidence intervals. Dashed lines indicate chance performance. There was some variability in performance across the different trials. Notably, even adult participants had mixed intuitions about the glasses & pillow trial.

Supplemental Materials D

Experiment 3

Example trial

Figure D1 shows an an example trial from Experiment 3.

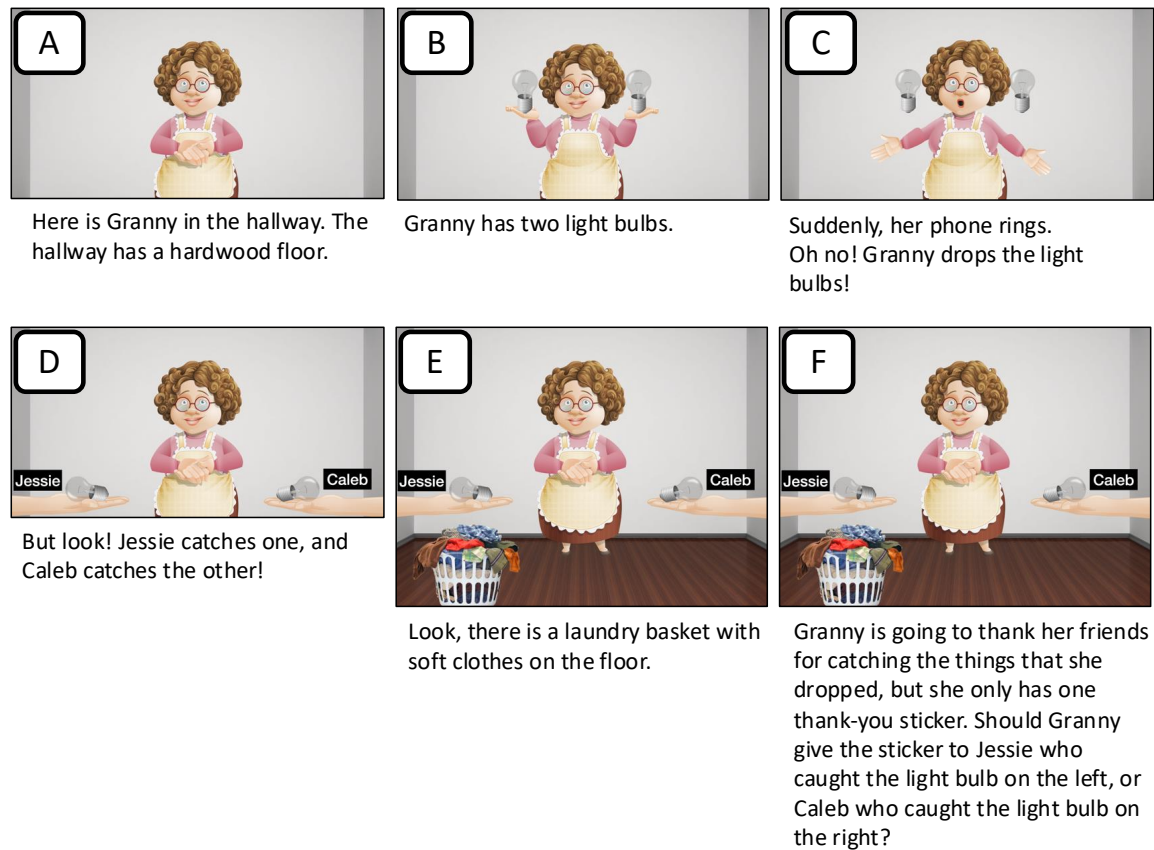


Figure D1

***Experiment 3 example trial:** Each panel shows what participants saw and what they were told. Notice that unlike in Experiment 2, participants didn't view the full scene at the beginning. The fact that there was a laundry basket underneath one of the light bulbs was only revealed after the light bulbs were caught. This means that participants could not anticipate what would happen if the light bulbs were dropped at the beginning of the scene, but instead had to consider what would have happened if the light bulbs hadn't been caught at the end of the scene.*

Trial results

The results for each of the six trials are shown below. As can be seen in [Figure D2](#), there was some variability in performance across the trials. For example, even adult participants were less certain about the 'glasses & pillow' trial. One possible reason is that while the pillow might prevent the glass from breaking, it would arguably be worse if the glass broke on the pillow rather than on the ground.

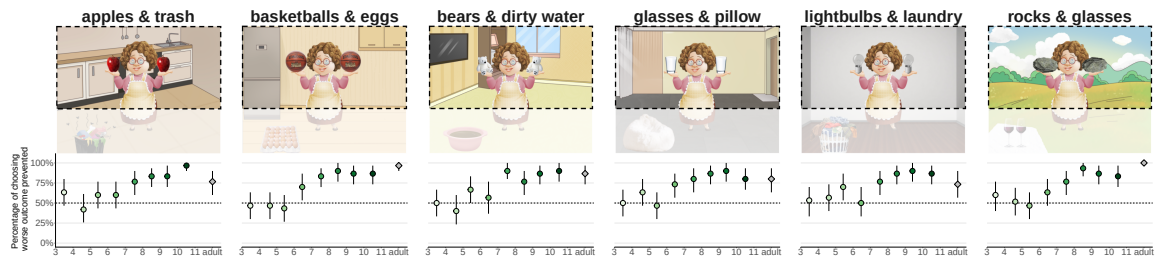


Figure D2

Experiment 3: Percentage of participants who chose the person who prevented the worse outcome from happening in each of the six test trials. Points show means for each age group with 95% bootstrapped confidence intervals. Dashed lines indicate chance performance. Children aged 6 years and older performed above chance overall. The pattern of performance was largely similar across the trials, with a notable increase in performance around age 7.